Balaji Balachandran

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Education

Stanford University GPA: 3.89/4.0

Master of Science in Computer Science w/ specialization in Network and Computer Security Sept. 2025 – June 2027

Relevant Coursework: Advanced Networking and Distributed Systems, Cryptography, Cryptocurrencies and Blockchains, Quantum Cryptography, Randomized Algorithms

University of Illinois Urbana Champaign GPA: 4.0/4.0

Bachelors of Science in Computer Science Aug. 2022 – May 2025

Relevant Coursework: Computer Security, Cryptography, Distributed Systems, Applied Parallel, Programming, Artificial Intelligence, Applied Machine Learning, Algorithms, Operating Systems, Networking, Linear Algebra, Computer Architecture, Internet of Things

Institutional Honors: University Honors, Bronze Tablet Scholar (highest UG honor), Dean's List 2022-2025

Experience

Empirical Security Research Group Oct 2025 – Present

Research Assistant, PI: Zakir Durumeric *Stanford, CA*

- Designing systems to improve efficiency of large-scale web measurement by reducing network transfers in automated crawlers.
- Built tooling to omit non-critical web resources, reducing bandwidth by 50% for 60% of sites while preserving fidelity.
- Formulated a novel modality-aware framework to classify web resources and evaluate network efficiency vs. site fidelity
- Implemented shared-socket architecture in Zgrab for multiplexed scanning from a single IP/port, improving connection reuse.
- Coauthored *"Stop Trying To Make Fetch Happen": Towards a Modality-Aware Understanding of Unnecessary Resources on the Web* to appear in IMC '26

LinkedIn June 2026 – Present

System and Infrastructure Engineer Intern *Mountain View, CA*

- Architected and independently built a dedicated microservice by decoupling traffic validation from disaster recovery, enabling continuous service deployment and isolated iteration.
- Implemented a Kalman filter algorithm and statistical analysis models to detect pre-failover anomalies and guarantee data robustness to mitigate failure during traffic shifts.
- Engineered a declarative simulation-testing framework to validate algorithm modifications offline, bypassing production risk and significantly accelerating feature development.

SysNet Lab May 2024 – Sept 2024

Research Assistant, PI: Radhika Mittal *Champaign IL*

- Used eBPF to trace requests at kernel-level across distributed microservices to enable observability without modifying code.
- Developed probabilistic techniques to associate microservice calls with requests across various concurrent threading models.
- Built a passive tracing model that reconstructs end-to-end request paths across services using observed network and execution patterns.

SysNet Lab May 2024 – Sept 2024

Research Assistant, PI: Radhika Mittal *Champaign IL*

- Used eBPF to trace requests at kernel-level across distributed microservices to enable observability without modifying code.
- Developed probabilistic techniques to associate microservice calls with requests across various concurrent threading models.
- Built a passive tracing model that reconstructs end-to-end request paths across services using observed network and execution patterns.

Teaching

Stanford University Sept 2026 – Dec 2026

Head Course Assistant, Randomized Algorithms and Probabilistic Analysis [CS 265] *Stanford, CA*

- Held office hours and supported graduate and advanced undergraduate students in probabilistic analysis, concentration bounds, and the design of randomized algorithms.

University of Illinois Urbana-Champaign Jan 2024 – May 2025

Course Assistant, Prob & Stats in CS [CS 361] and Algorithms [CS 374] *Champaign, IL*

- Conducted weekly office hours for two courses with 350+ students per semester.
- Supported two of the largest CS courses required of every undergraduate, helping students apply statistics concepts to computer science and methodically design algorithms.

Projects

Communipute | *Python, React, Typescript, Flask, Docker, Kubernetes, JWT* February 2023

- Developed a crowd-sourced cloud service allowing users to donate and request additional compute power from their community. Implemented secure authentication through JWT, networking through polling, and a React+TypeScript frontend
- Winner of three track prizes at Stanford's TreeHacks 2023 hackathon (the world's largest collegiate hackathon with **1600+** participants)

Technical Skills & Honors

Languages: C/C++, Go, Python, Solidity, Bash, Swift, Kotlin, JavaScript, Typescript, HTML/CSS, MySQL
Frameworks: React, Node.js, Flask, React Native, Microsoft Azure, FastAPI, Selenium, CUDA, AWS, Koin
Honors: Winner of Stanford's Treehacks (2023), UIUC's HackIllinois (2023), and Synchrony's hackathon (2023)